Fix-round CMSSW emu, RelVal file 3 (test events): residuals vs the 1.5x target; class-C route closure

CMS alpha (bending) - IL2-4

CMS alpha (bending) - IL1

CMS alpha (bending) - ID1-4

density (unit area)

residual [deg] (ends = overflow)

CMS beta - IL2-4

CMS beta - IL1

CMS beta - ID1-4

SmartPixels dump 1x (PixelAV angle mix)

TARGET: dump, bias kept + 1.5x spread, PixelAV angle mix

TARGET reweighted to the class-T Phase-2 angles (the class-T reference)

CMSSW emu class T: emitted - true

class-C route on class-T clusters (CMSSW CNN + A1 draws; closure)

CMSSW emu class C linked: emitted - (inconsistent) TP truth

Residual statistics, CMS alpha [deg] (core = mode-seeded Gaussian core RMS about 0; full = RMS about 0)

group	curve	core_rms	full_rms	frac_gt10	N	fast_study_full_rms
IL2-4	SmartPixels dump 1x (PixelAV angle mix)	1.273	3.783	0.0157	108222	
IL2-4	TARGET: dump, bias kept + 1.5x spread, PixelAV angle mix	1.855	5.643	0.03059	108222	
IL2-4	TARGET reweighted to the class-T Phase-2 angles (the class-T reference)	4.363	6.895	0.06815	108222	
IL2-4	CMSSW emu class T: emitted - true	4.317	6.911	0.06824	1174067	6.9
IL2-4	class-C route on class-T clusters (CMSSW CNN + A1 draws; closure)	7.041	12.73	0.2368	235428	12.87
IL2-4	CMSSW emu class C linked: emitted - (inconsistent) TP truth	39.4	44.5	0.7295	409184	
IL1	SmartPixels dump 1x (PixelAV angle mix)	1.273	3.783	0.0157	108222	
IL1	TARGET: dump, bias kept + 1.5x spread, PixelAV angle mix	1.855	5.643	0.03059	108222	
IL1	TARGET reweighted to the class-T Phase-2 angles (the class-T reference)	3.623	6.599	0.0562	108222	
IL1	CMSSW emu class T: emitted - true	3.55	6.59	0.05616	719916	6.57
IL1	class-C route on class-T clusters (CMSSW CNN + A1 draws; closure)	5.633	11.71	0.148	144753	11.47
IL1	CMSSW emu class C linked: emitted - (inconsistent) TP truth	46.66	46.25	0.7672	83606	
ID1-4	SmartPixels dump 1x (PixelAV angle mix)	1.273	3.783	0.0157	108222	
ID1-4	TARGET: dump, bias kept + 1.5x spread, PixelAV angle mix	1.855	5.643	0.03059	108222	
ID1-4	TARGET reweighted to the class-T Phase-2 angles (the class-T reference)	4.384	6.293	0.04771	108222	
ID1-4	CMSSW emu class T: emitted - true	4.554	6.58	0.04943	3229453	6.59
ID1-4	class-C route on class-T clusters (CMSSW CNN + A1 draws; closure)	5.608	9.152	0.1211	648524	9.15
ID1-4	CMSSW emu class C linked: emitted - (inconsistent) TP truth	10.11	38.67	0.5678	476705	

Residual statistics, CMS beta [deg] (core = mode-seeded Gaussian core RMS about 0; full = RMS about 0)

group	curve	core_rms	full_rms	frac_gt10	N	fast_study_full_rms
IL2-4	SmartPixels dump 1x (PixelAV angle mix)	0.4863	6.407	0.06576	108222	
IL2-4	TARGET: dump, bias kept + 1.5x spread, PixelAV angle mix	0.7196	9.435	0.09433	108222	
IL2-4	TARGET reweighted to the class-T Phase-2 angles (the class-T reference)	5.049	15.14	0.3144	108222	
IL2-4	CMSSW emu class T: emitted - true	5.056	15.12	0.3146	1174067	15.08
IL2-4	class-C route on class-T clusters (CMSSW CNN + A1 draws; closure)	9.332	23.04	0.4574	235428	23.26
IL2-4	CMSSW emu class C linked: emitted - (inconsistent) TP truth	44.44	53.12	0.7894	409184	
IL1	SmartPixels dump 1x (PixelAV angle mix)	0.4863	6.407	0.06576	108222	
IL1	TARGET: dump, bias kept + 1.5x spread, PixelAV angle mix	0.7196	9.435	0.09433	108222	
IL1	TARGET reweighted to the class-T Phase-2 angles (the class-T reference)	0.8335	11.91	0.1736	108222	
IL1	CMSSW emu class T: emitted - true	0.8024	11.83	0.1726	719916	11.79
IL1	class-C route on class-T clusters (CMSSW CNN + A1 draws; closure)	1.998	29.87	0.332	144753	30.21
IL1	CMSSW emu class C linked: emitted - (inconsistent) TP truth	55.33	66.01	0.7606	83606	
ID1-4	SmartPixels dump 1x (PixelAV angle mix)	0.4863	6.407	0.06576	108222	
ID1-4	TARGET: dump, bias kept + 1.5x spread, PixelAV angle mix	0.7196	9.435	0.09433	108222	
ID1-4	TARGET reweighted to the class-T Phase-2 angles (the class-T reference)	4.893	17.99	0.6226	108222	
ID1-4	CMSSW emu class T: emitted - true	19.4	18.25	0.6258	3229453	18.25
ID1-4	class-C route on class-T clusters (CMSSW CNN + A1 draws; closure)	19.79	19.25	0.6365	648524	19.28
ID1-4	CMSSW emu class C linked: emitted - (inconsistent) TP truth	30.6	47.25	0.7758	476705	